

MTH 204 : Lecture 25

Last time we looked at D'Alembert's solution of the wave equation: $u_{tt} = c^2 u_{xx}$ ($c^2 = \frac{p}{\rho}$)

By changing the variable $v = x + ct$
and $w = x - ct$, the wave equation becomes $u_{vw} = 0$

- Integrating with respect to v and w we get $u = \phi(x + ct) + \psi(x - ct)$
(Sum of two possibly different travelling waves)
- After this we have used the initial conditions to obtain the solution u in terms of the functions given in the initial conditions.

Method of characteristics:

The idea of d'Alembert's solution is a special case of method of characteristics that deals with the problem:

$$A u_{xx} + 2B u_{xy} + C u_{yy} = F(x, y, u, u_x, u_y)$$

The Equation is quasilinear (i.e. linear in highest derivatives)
Depending on the discriminant $AC-B^2$, we can classify:

<u>Type</u>	<u>Defining Condition</u>	<u>Example</u>
Hyperbolic	$AC-B^2 < 0$	Wave Equation
Parabolic	$AC-B^2 = 0$	Heat Equation
Elliptic	$AC-B^2 > 0$	Laplace Equation

A, B, C may be functions of x and y .
So, the problem may be of mixed type
that is different type in different regions

Ex: Consider the 1D wave equation:

$$u_{tt} = c^2 u_{xx}$$

Making a change of variable $y = ct$,

We have $u_{tt} = c^2 u_{yy}$

So, the wave equation becomes $c^2 u_{yy} = c^2 u_{xx}$

$$\Rightarrow \boxed{u_{xx} - u_{yy} = 0} \quad (A=1, B=0, C=-1)$$

Now, $AC - B^2 = (1)(-1) - 0^2 = -1 < 0$

Equation is hyperbolic.

Ex: Consider the 1D heat equation

$$u_t = c^2 u_{xx}$$

Now making the change of variable $y = c^2 t$,
we get $u_t = c^2 u_y$

Then the equation becomes $u_{xx} = u_y$ ($A=1, B=0, C=0$)

$$\text{Then } AC - B^2 = 1(0) - 0^2 = 0$$

So, the equation is parabolic.

Transformation to Normal form

The characteristic Equation of the PDE

is the ODE $Ay'^2 - 2By' + C = 0$

where $y' = \frac{dy}{dx}$

- The normal forms of the PDE and the corresponding transformations depend on the type of PDE.

They are obtained by solving the characteristic equation.

The solutions of the characteristic equation are called the characteristics of the PDE and are written in the form $\psi(x, y) = \text{constant}$, $\Phi(x, y) = \text{constant}$

Then the transformations giving new variables v, w instead of x, y and the normal forms of the PDE can be given:

<u>Type</u>	<u>New Variables</u>	<u>Normal Form</u>
Hyperbolic	$v = \Phi, w = \psi$	$\mathcal{U}_{vw} = F_1$
Parabolic	$v = x, w = \Phi = \psi$	$\mathcal{U}_{ww} = F_2$
Elliptic	$v = \frac{1}{2}(\Phi + \psi), w = \frac{1}{2i}(\Phi - \psi)$	$\mathcal{U}_{vv} + \mathcal{U}_{ww} = F_3$

D'Alembert Solution

$$u_{tt} - c^2 u_{xx} = 0$$

With the change of variable $y = ct$, the above PDE is transformed into

$$u_{xx} - u_{yy} = 0 \quad (A=1, C=-1, B=0)$$

The characteristic equation is

$$(y')^2 - 1 = 0$$

$$\Rightarrow (y' + 1)(y' - 1) = 0$$

$$\Rightarrow y' + 1 = 0 \Rightarrow y' = -1 \Rightarrow y = -x + C_1$$

$$\Rightarrow x + y = C_1$$

$$\Rightarrow \Phi(x, y) = x + y = C_1$$

$$\text{and } y' - 1 = 0 \Rightarrow y' = 1 \Rightarrow y = x + C_2'$$

$$\Rightarrow x - y = C_2 \quad (C_2 = -C_2')$$

$$\Rightarrow \Psi(x, y) = x - y = C_2$$

Since the equation is hyperbolic, the change of variable is

$$v = \Phi(x, y) = x + y = x + ct$$

$$w = \Psi(x, y) = x - y = x - ct$$

and the associated normal form is:

$$u_{vw} = 0$$

Now we can solve this equation as before.

Ex: $u_{xx} + u_{xy} - 2u_{yy} = 0$

So, $A = 1$, $B = \frac{1}{2}$, $C = -2$

Hence $AC - B^2 = 1(-2) - \left(\frac{1}{2}\right)^2 = -\frac{9}{4} < 0$

So, it is a Hyperbolic Equation.

The characteristic Equation is: $y'^2 - 2\frac{1}{2}y' - 2 = 0$

$$\Rightarrow y'^2 - y' - 2 = 0 \Rightarrow (y' - 2)(y' + 1) = 0$$

$$\Rightarrow y' + 1 = 0 \Rightarrow y = -x + c_1 \Rightarrow x + y = c_1 \Rightarrow \Phi(x, y) = x + y = c_1$$

and $y' - 2 = 0 \Rightarrow y = 2x + c_2' \Rightarrow 2x - y = c_2$ (where $c_2 = -c_2'$)

$$\Rightarrow \Psi(x, y) = 2x - y = c_2$$

Then the change of variable will be :

$$v = \Phi(x, y) = x + y$$

and $w = \Psi(x, y) = 2x - y$

So, $u_x = u_v v_x + u_w w_x = u_v(1) + u_w(2) = u_v + 2u_w$

Then $u_{xx} = (u_v + 2u_w)_v v_x + (u_v + 2u_w)_w w_x$

$$= u_{vv} + 2u_{wv} + 2u_{vw} + 4u_{ww}$$

$$= u_{vv} + 4u_{vw} + 4u_{ww}$$

$$u_{xy} = (u_v + 2u_w)_v v_y + (u_v + 2u_w)_w w_y$$

$$= u_{vv} + 2u_{wv} - u_{vw} - 2u_{ww}$$

$$= u_{vv} + u_{vw} - 2u_{ww}$$

Also $u_y = u_v v_y + u_w w_y = u_v - u_w$

$$\text{Now } u_{yy} = (u_v - u_w)_v v_y + (u_v - u_w)_w w_y$$

$$= u_{vv} - u_{wv} - u_{vw} + u_{ww}$$

$$= u_{vv} - 2u_{vw} + u_{ww}$$

$$\text{So, the equation } u_{xx} + u_{xy} - 2u_{yy} = 0$$

$$\text{becomes } \cancel{u_{vv}} + 4u_{vw} + \cancel{u_{ww}} + \cancel{u_{vv}} + u_{vw} - 2\cancel{u_{ww}}$$

$$- 2\cancel{u_{vv}} + 4u_{vw} - 2\cancel{u_{ww}} = 0$$

$$\Rightarrow 9u_{vw} = 0$$

$$\Rightarrow \boxed{u_{vw} = 0}$$

Integrating with respect to w

$$u_v = f(v) \quad (\text{a constant with respect to } w)$$

Integrating with respect to v ,

$$u = \int f(v) dv$$

$$\Rightarrow u = f_1(v) + f_2(w) \quad (\text{a constant with respect to } v)$$

$$\Rightarrow \boxed{u = f_1(x+y) + f_2(2x-y)}$$

1D Heat Equation:



(Bar under consideration)

The PDE is $u_t = c^2 u_{xx}$

with the boundary conditions:

$$u(0, t) = 0, \quad u(L, t) = 0 \quad \text{for all } t \geq 0$$

(i.e. the two ends of the bar are kept at temperature 0)

with the initial condition $u(x, 0) = f(x)$ (given)

(By above $f(0) = 0, f(L) = 0$)

The solution has three steps: (a) Separating variables

(b) Satisfying the boundary conditions

(c) Satisfying the initial condition

(a) Let us try a solution of the form

$$u(x, t) = F(x) G(t)$$

Then $u_t = F G_t$, $u_x = F_x G$, $u_{xx} = F_{xx} G$

Substituting we get $F G_t = c^2 F_{xx} G$

$$\Rightarrow \frac{G_t}{c^2 G} = \frac{F_{xx}}{F}$$

The left hand side depends only on t and the right hand side depends only on x .

So, it must be

$$\frac{G_t}{c^2 G} = \frac{F_{xx}}{F} = \text{constant } k$$

- We can show that for $k=0$ or $k>0$, the only solution $u = FG$ satisfying the boundary conditions is $u=0$

- Therefore $\frac{G_t}{c^2 G} = \frac{F_{xx}}{F} = -p^2$

This gives us the two equations

$$\left. \begin{aligned} F_{xx} + p^2 F &= 0 \\ G_t + c^2 p^2 G &= 0 \end{aligned} \right\}$$

Now $F_{xx} + p^2 F = 0 \Rightarrow F = A_1 \cos(px) + B_1 \sin(px)$

Now $G_t + c^2 p^2 G \Rightarrow \frac{G_t}{G} = -c^2 p^2$
 $\Rightarrow \ln|G| = -c^2 p^2 t + C_1 \Rightarrow G = C_2 e^{-c^2 p^2 t}$

Hence $u(x,t) = [A \cos(px) + B \sin(px)] e^{-c^2 p^2 t}$

(b) Boundary conditions:

$$u(0,t) = 0 \Rightarrow A e^{-c^2 p^2 t} = 0 \Rightarrow A = 0$$

$$u(L,t) = 0 \Rightarrow B(\sin pL) e^{-c^2 p^2 t} = 0$$

$$\Rightarrow B \sin(pL) = 0 \Rightarrow pL = n\pi$$

Let us define $\lambda_n = c \frac{n\pi}{L}$

Then the eigen functions of the problem

$$u_n(x,t) = B_n \sin\left(\frac{n\pi}{L} x\right) e^{-\lambda_n^2 t}$$

and $\lambda_n = c \frac{n\pi}{L}$ is the corresponding eigen values.

The solution of the problem is

$$u(x,t) = \sum_{n=1}^{\infty} u_n(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{L} x\right) e^{-\lambda_n^2 t}$$

Now the initial condition will give

$$u(x, 0) = f(x) \Rightarrow \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{L} x\right) = f(x)$$

So, B_n must be the coefficients of the sine Fourier series

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$$

for $n=1, 2, \dots$

Because of the exponential factor,
all the terms in the solution approaches
zero as t approaches infinity.
The rate of decay increases with n .

Example (Sinusoidal Initial temperature):

Find the temperature $u(x, t)$ in a laterally insulated copper bar 80 cm long if the initial temperature is $100 \sin(\pi x / 80)^\circ\text{C}$ and the ends are kept at 0°C .

How long will it take for the maximum temperature in the bar to drop to 50°C ?

Physical data for Copper: density 8.92 g/cm^3
Specific heat $0.092 \text{ cal/(g}^\circ\text{C)}$
Thermal conductivity $0.95 \text{ cal/(cm sec}^\circ\text{C)}$

Now from the problem $f(x) = 100 \sin\left(\frac{\pi}{80} x\right)$
 $\Rightarrow B_1 = 100$, $B_n = 0$ for $n = 2, 3, \dots$

$$\text{Now } \lambda_1^2 = \frac{c^2 \pi^2}{L^2}$$

$$c^2 = \frac{k}{\sigma \rho} = \frac{.95}{.092 \times 8.92} = 1.158 \left(\frac{\text{cm}^2}{\text{s}} \right)$$

$$\begin{aligned} \lambda_1^2 &= c^2 \frac{\pi^2}{L^2} = 1.158 \left(\frac{\text{cm}^2}{\text{s}} \right) \cdot \frac{\pi^2}{(80)^2 (\text{cm}^2)} \\ &= 1.785 \times 10^{-3} (\text{s}^{-1}) \end{aligned}$$

Thus we get the solution as:

$$u(x, t) = 100 \sin\left(\frac{\pi}{80} x\right) e^{-1.785 \times 10^{-3} t}$$

To calculate the time for the maximum }
temperature to drop to 50°C , }

$$100 e^{-1.785 \times 10^{-3} t} = 50$$

$$\Rightarrow t = \frac{\log(0.5)}{-1.785 \times 10^{-3}} = 388 \text{ second}$$

$$= 6 \text{ minute } 28 \text{ seconds}$$

$$\approx 6.5 \text{ minutes.}$$

Ex: Solve the same problem with initial temperature $f(x) = 100 \sin\left(\frac{3\pi x}{80}\right)^\circ\text{C}$ and the other data as before.

• Instead of $n=1$, we now have $n=3$

$$B_3 = 100, \quad B_n = 0 \text{ for } n=1, 2, 4, 5, \dots$$

$$\lambda_3^2 = 3^2 \lambda_1^2 = 9 \times 1.785 \times 10^{-3} = 1.607 \times 10^{-2}$$

So, the solution is

$$u(x, t) = 100 \sin\left(\frac{3\pi x}{80}\right) e^{-1.607 \times 10^{-2} t}$$

For the maximum temperature to drop to 50°C ,

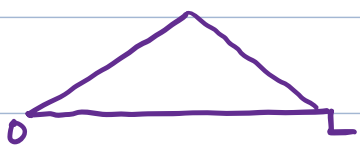
$$100 e^{-1.607 \times 10^{-2} t} = 50$$

$$\Rightarrow t = \frac{\log(0.5)}{-1.607 \times 10^{-2}} = 43 \text{ seconds.}$$

(Approximately 9 times as fast as the previous example)

Ex: Triangular Initial temperature in a Bar:

Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0, assuming that the initial temperature is



$$f(x) = \left. \begin{array}{ll} x & \text{if } 0 \leq x \leq \frac{L}{2} \\ L-x & \text{if } \frac{L}{2} \leq x \leq L \end{array} \right\}$$

• Here we will get

$$B_n = \frac{2}{L} \left(\int_0^{L/2} x \sin \frac{n\pi}{L} x \, dx + \int_{L/2}^L (L-x) \sin \frac{n\pi}{L} x \, dx \right)$$

Then $B_n = 0$ if n is even

$$\text{and } B_n = \frac{4L}{n^2 \pi^2} \quad (\text{for } n=1, 5, 9, \dots)$$

$$\text{and } B_n = -\frac{4L}{n^2 \pi^2} \quad (\text{for } n=3, 7, 11, \dots)$$

Hence, the solution is:

$$u(x, t) = \frac{4L}{\pi^2} \left[\sin\left(\frac{\pi x}{L}\right) e^{-\left(\frac{c\pi}{L}\right)^2 t} - \frac{1}{9} \sin\left(\frac{3\pi x}{L}\right) e^{-\left(\frac{3c\pi}{L}\right)^2 t} + \dots \right]$$

The temperature decreasing with increasing t because of heat loss due to the cooling of the ends.

Ex: Bar with insulated ends:

Find a solution of the problem where the boundary condition is replaced by the condition that both ends of the bar are insulated.

- The rate of heat flow is proportional to the gradient of the temperature.

Thus if the ends $x=0$ and $x=L$ of the bar are insulated so the no heat can flow through the ends, we have

$$u_x(0, t) = 0, \quad u_x(L, t) = 0 \quad \text{for all } t > 0$$

The equation and the initial conditions

Remain the same.
$$\left. \begin{aligned} u_t &= c^2 u_{xx} \\ u(x, 0) &= f(x) \end{aligned} \right\}$$

Since the equation has not changed, the solution is still of the form

$$u(x, t) = [A \cos(px) + B \sin(px)] e^{-c^2 p^2 t}$$

Now $u_x(x, t) = F_x(x) G(t)$

$$= [-Ap \sin(px) + Bp \cos(px)] e^{-c^2 p^2 t}$$

Now $u_x(0, t) = 0 \Rightarrow Bp = 0$

If $p=0$, the number of solutions is limited.

So, we choose $B=0$

Now $u_x(L, t) = 0 \Rightarrow -Ap \sin(pL) = 0$

$$\Rightarrow pL = n\pi \Rightarrow p = \frac{n\pi}{L}$$

Let $p_n = \frac{n\pi}{L}$

Then we have the eigen functions

$$u_n(x, t) = A_n \cos(p_n x) e^{-c^2 p_n^2 t} = A_n \cos\left(\frac{n\pi}{L} x\right) e^{-\lambda_n^2 t}$$

where $\lambda_n = \frac{c n \pi}{L}$

Note that now $n=0, 1, 2, \dots$

instead of $n=1, 2, \dots$

i.e. we can have the solution $u_0 = A_0$

(Thus 0 can be an eigen value)

$$\begin{aligned} u_n(x, t) &= A_n \cos(p_n x) e^{-c^2 p_n^2 t} \\ &= A_n \cos\left(\frac{n\pi}{L} x\right) e^{-\lambda_n^2 t} \end{aligned}$$

where $\lambda_n = \frac{c n \pi}{L}$

The particular solution must be of the form

$$u(x, t) = \sum_{n=0}^{\infty} u_n(x, t) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi}{L} x\right) e^{-\lambda_n^2 t}$$

Now the initial condition gives:

$$u(x, 0) = f(x) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi}{L} x\right)$$

So, the coefficients A_n are the coefficients of the Fourier cosine series of $f(x)$.

$$A_0 = \frac{1}{L} \int_0^L f(x) dx, \quad A_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi}{L} x\right) dx$$

Ex: A laterally insulated bar of length 10 cm and constant cross sectional area 1 cm^2 of density $\rho = 10.6 \text{ gm/cm}^3$, thermal conductivity $k = 1.04 \text{ cal/(cm sec}^\circ\text{C)}$ and specific heat $\sigma = 0.056 \text{ cal/(gm}^\circ\text{C)}$. (This corresponds to silver) has initial temperature $f(x)$ and is kept at 0°C at the ends $x=0$ and $x=10$. Find the temperature $u(x, t)$ at later times.

$$f(x) = \sin(0.1\pi x) + \frac{1}{2} \sin(0.2\pi x)$$

Here $c^2 = \frac{k}{\sigma \rho} = \frac{1.04}{0.056 \times 10.6} = 1.752$

Since $u(0, t) = 0$, $u(10, t) = 0 \quad \forall t \geq 0$ ($L = 10 \text{ cm}$), the solution of the heat equation is:

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\lambda_n^2 t}$$

Now $L = 10$

$$\text{So, } \lambda_n = \frac{cn\pi}{L} \Rightarrow \lambda_n^2 = \frac{c^2 n^2 \pi^2}{L^2} = \frac{1.752 n^2 \pi^2}{100}$$

Now $u(x, 0) = f(x)$

$$\Rightarrow \sum_{n=1}^{\infty} B_n \left(\sin \frac{n\pi x}{L} \right) = f(x) = \sin(0.1\pi x) + \frac{1}{2} \sin(0.2\pi x)$$

clearly $B_n = 0 \quad \forall n > 3$

and $B_1 \sin\left(\frac{\pi}{10}x\right) + B_2 \sin\left(\frac{2\pi}{10}x\right) = \sin(0.1\pi x) + \frac{1}{2} \sin(0.2\pi x)$

$$\Rightarrow B_1 = 1 \quad \text{and} \quad B_2 = \frac{1}{2}$$

So,
$$u(x,t) = \sin(0.1\pi x) e^{-\frac{1.752}{100} \pi^2 t} + \frac{1}{2} \sin(0.2\pi x) e^{-\frac{1.752}{100} 4\pi^2 t}$$



Steady two dimensional Heat Problems:

Consider the two dimensional heat equation

$$\frac{\partial u}{\partial t} = c^2 \nabla^2 u = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

For steady state (i.e. time independent)

$$\frac{\partial u}{\partial t} = 0$$

and so heat equation becomes

$$\nabla^2 u = 0$$

This equation will be considered in some region

R of the xy -plane and a boundary condition

on the boundary curve C of R is given.

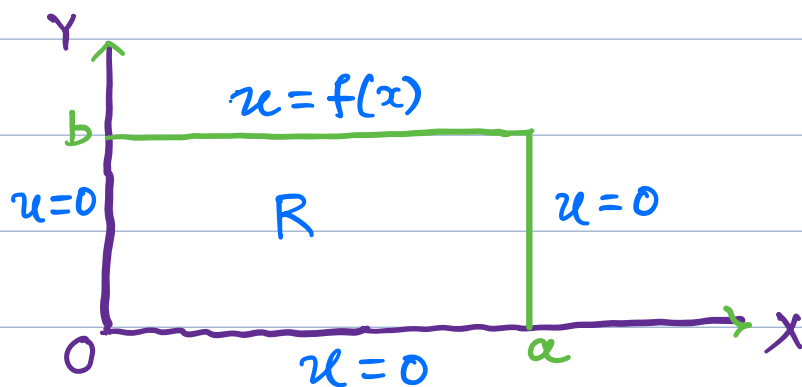
Thus this is a BVP (Boundary value problem). This will be called:

(1) Dirichlet Problem: If u is prescribed on C

(2) Neumann Problem: If normal derivative
 $u_n = \frac{\partial u}{\partial n}$ is prescribed on C

(3) Robin Problem: If u is prescribed on some portion of C and u_n on the rest of C (Mixed Boundary Condition)

Dirichlet Problem in a Rectangle R



Let us solve the problem by separation of variables:

$$\text{Let } u(x, y) = F(x)G(y)$$

$$\text{Then } \nabla^2 u = 0 \Rightarrow F_{xx}G + FG_{yy} = 0$$

$$\Rightarrow \frac{F_{xx}}{F} = -\frac{G_{yy}}{G} = -k$$

$$\Rightarrow F_{xx} + kF = 0 \Rightarrow F = A \cos(\sqrt{k}x) + B \sin(\sqrt{k}x)$$

$$\text{Now } F(0) = 0 \Rightarrow A(1) = 0 \Rightarrow A = 0$$

$$F(a) = 0 \Rightarrow B \sin(\sqrt{k}a) = 0 \Rightarrow \sqrt{k}a = n\pi$$
$$\Rightarrow k = \left(\frac{n\pi}{a}\right)^2$$

The non zero solutions are $F_n(x) = \sin\left(\frac{n\pi}{a}x\right)$
for $n=1, 2, \dots$

Now $G_{yy} - kG = 0$

$$\Rightarrow G_n = A_n e^{\sqrt{k} y} + B_n e^{-\sqrt{k} y} \quad \text{where } k = \left(\frac{n\pi}{a}\right)^2$$

$$\Rightarrow G_n = A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}}$$

Now the boundary conditions

$$G_n(0) = 0 \Rightarrow A_n + B_n = 0 \Rightarrow B_n = -A_n$$

$$\begin{aligned} \text{Then } G_n &= A_n e^{\frac{n\pi y}{a}} - A_n e^{-\frac{n\pi y}{a}} \\ &= 2A_n \sinh\left(\frac{n\pi y}{a}\right) \end{aligned}$$

Therefore the eigen functions are

$$u_n(x, y) = F_n(x) G_n(y) = A_n^* \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right)$$

$$\text{The solution is } u(x, y) = \sum_{n=1}^{\infty} u_n(x, y) = \sum_{n=1}^{\infty} A_n^* \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right)$$

Now $u(x, b) = f(x)$

$$\Rightarrow \sum_{n=1}^{\infty} A_n^* \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi b}{a}\right) = f(x)$$

$$\Rightarrow f(x) = \sum_{n=1}^{\infty} \left[A_n^* \sinh\left(\frac{n\pi b}{a}\right) \right] \sin\left(\frac{n\pi x}{a}\right)$$

Hence $A_n^* \sinh\left(\frac{n\pi b}{a}\right)$ must be the Fourier coefficients b_n of $f(x)$

$$\text{So, } b_n = A_n^* \sinh\left(\frac{n\pi b}{a}\right) = \frac{2}{a} \int_0^a f(x) \sin\left(\frac{n\pi x}{a}\right) dx$$

Therefore the solution of the problem is:

$$u(x, y) = \sum_{n=1}^{\infty} u_n(x, y) = \sum_{n=1}^{\infty} A_n^* \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right)$$

$$\text{where } A_n^* = \frac{2}{a \sinh\left(\frac{n\pi b}{a}\right)} \int_0^a f(x) \sin\left(\frac{n\pi x}{a}\right) dx$$